Parameter change files

Purpose

`cal_parms.cal` is the list of parameters that calibration may change: the name of each parameter, the type of object it belongs to, and the lowest and highest value it may take. `calibration.cal` is the change list for one run: each row picks a parameter from `cal_parms.cal`, says how to change it, and optionally limits the change to some objects or to objects that meet conditions. How SWAT+ applies the list is described on Hard calibration.

Both files are named on the `chg` line of `file.cio`, in the first two positions (defaults `cal_parms.cal` and `calibration.cal`). A `null` name or a missing file switches the change list off.

All values are free format and separated by blanks. Names must not contain blanks.

cal_parms.cal

- Line 1: title (skipped).
- Line 2: number of parameter rows.
- Line 3: column header (skipped).
- Then one row per parameter.

#	Column	Description
1	`name`	Parameter name, as listed under Parameter names.
2	`obj_typ`	Type of object the parameter belongs to: `hru`, `sol`, `plt`, `hlt`, `aqu`, `sdc`, `rte`, `swq`, `res`, `rdt`, `bsn`, `pst`, `gwf`, `gwf_riv`, `gwf_sgl` or `cli`. It decides which objects are changed when a row of `calibration.cal` lists none.
3	`abs_min`	Lowest value the parameter may take. Every change is limited to this value.
4	`abs_max`	Highest value the parameter may take. Every change is limited to this value.
5	`units`	Units, for information only.

SWAT+ Editor writes this file with suitable limits for each parameter, and it rarely needs editing. A narrower range for a calibration run belongs in the calibration tool; the range here is the physical limit.

Example from `refdata/Osu_1hru` in the SWAT+ repository:

```
cal_parms.cal: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
186
name                      obj_typ       abs_min       abs_max         units
cn2                           hru      35.00000      95.00000          null
cn3_swf                       hru       0.00000       1.00000          null
usle_p                        hru       0.00000       1.00000          null
ovn                           hru       0.01000      30.00000          null
elev                          hru       0.00000    5000.00000             m
slope                         hru       0.00010       0.90000           m/m
slope_len                     hru      10.00000     150.00000             m
lat_ttime                     hru       0.50000     180.00000          days
```

The count on line 2 must match the number of rows. Rows past the count are not read.

calibration.cal

- Line 1: title (skipped).
- Line 2: number of change rows.
- Line 3: column header (skipped).
- Then one row per change, each followed directly by its condition lines, if it has any.

Change rows

#	Column	Description
1	`name`	Parameter name. Must also be in `cal_parms.cal`.
2	`chg_typ`	`absval` (replace), `abschg` (add), `pctchg` (percent change) or `relchg` (relative change). See Change types.
3	`val`	Size of the change.
4	`conds`	Number of condition lines that follow this row. 0 for none.
5	`lyr1`	First soil layer to change, for `sol` parameters (0 = top layer). For reservoir decision tables, a zone switch.
6	`lyr2`	Last soil layer to change, for `sol` parameters (0 = bottom layer). For reservoir decision tables, a zone switch.
7	`year1`	First year of the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
8	`year2`	Last year of the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
9	`day1`	First day of the year in the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
10	`day2`	Last day of the year in the window, for `precip` and `temp`.
11	`obj_tot`	Number of entries in the object list that follows on the same line. 0 means every object of the parameter's type.
12…	object list	`obj_tot` object numbers.

The object numbers are the sequence numbers of the objects (the first HRU is 1, and so on). A negative number closes a range started by the number before it, so `obj_tot` = 3 with the list `1 -20 35` means objects 1 to 20 and object 35.

Condition lines

Each change row is followed by `conds` condition lines. A condition line has one of two forms:

- `range <var> <min> <max>`: the object's value of `<var>` must lie between `<min>` and `<max>`. Only `res_pvol` is recognised.
- `<var> <alt> <targ> <targc>`: a test against the text value `<targc>`. `<alt>` and `<targ>` must be present but are not used by the conditions that exist.

The variables and how they are tested are on Conditions and regionalization.

Example

`refdata/Osu_1hru/calibration.cal`:

```
  hru-new.cal developed from soft data calibration
    13
 NAME           CHG_TYP                 VAL    CONDS  LYR1  LYR2    YEAR1  YEAR2   DAY1   DAY2  OBJ_TOT
cn2          pctchg                7.714000E+00  0          0          0          0          0
petco        pctchg               -5.273950E+01  0          0          0          0          0
latq_co      pctchg               -3.732030E+01  0          0          0          0          0
alpha        pctchg                6.859000E+00  0          0          0          0          0
epco         pctchg                6.534100E+01  0          0          0          0          0
awc          pctchg               -6.256550E+01  0          0          0          0          0
esco         pctchg                3.305500E+01  0          0          0          0          0
perco        pctchg                1.434000E+00  0          0          0          0          0
```

```
cn3_swf     pctchg          -7.710000E-02  0  0  0  0  0
canmx       pctchg          -6.680230E+01  0  0  0  0  0
lat_len     pctchg          -6.241200E+00  0  0  0  0  0
revap_co    pctchg          -5.854530E+01  0  0  0  0  0
surlag      pctchg          -2.125720E+01  0  0  0  0  0
```

Every row has no conditions and an empty object list, so each change applies to all objects of its type. The rows have one more zero than the header has columns; the reader stops after `obj_tot` when it is 0, so the extra value is ignored.

A row that changes `cn2` by +5 only on HRUs 1 to 20 and HRU 35, and only where the hydrologic soil group is `C`:

```
cn2         abschg      5.0    1   0   0   0   0   0   0   3   1  -20   35
hsg         =           0      C
```

Parameter names

The names SWAT+ 62.0.1 recognises are listed below with the object type to give in `cal_parms.cal` and the input the parameter comes from. A name that is in neither list is skipped without a message. Some recognised names have no effect at 62.0.1, or change a different value; the notes with each name say which.

HRU parameters (`hru`)

Name	Input	Notes
`cn2`	condition II curve number from `cntable.lum`	The curve numbers for the other moisture conditions are recomputed.
`cn3_swf`	`cn3_swf`, `hydrology.hyd`	Not changed on tile-drained HRUs.
`perco`	`perco`, `hydrology.hyd`	Not changed on tile-drained HRUs.
`esco`, `epco`	`esco`, `epco`, `hydrology.hyd`	`epco` is also set for every plant in the HRU.
`petco`	`pet_co`, `hydrology.hyd`	
`latq_co`, `lat_sed`	`latq_co`, `lat_sed`, `hydrology.hyd`	
`lat_ttime`	`lat_ttime`, `hydrology.hyd`	Use `absval` with a value in days. `abschg`, `pctchg` and `relchg` act on an internal coefficient and give wrong travel times.
`canmx`	`can_max`, `hydrology.hyd`	
`biomix`	`bio_mix`, `hydrology.hyd`	
`erorgn`, `erorgp`	`orgn_enrich`, `orgp_enrich`, `hydrology.hyd`	
`lat_orgn`, `lat_orgp`	`lat_orgn`, `lat_orgp`, `hydrology.hyd`	
`slope`, `slope_len`, `lat_len`, `dis_stream`	`slp`, `slp_len`, `lat_len`, `dist_cha`, `topography.hyd`	
`elev`	HRU elevation	
`ovn`	Manning's n for overland flow, `ovn_table.lum`	
`usle_p`	USLE support practice factor, `cons_practice.lum`	
`field_len`, `field_wid`, `field_ang`	`len`, `wd`, `ang`, `field.fld`	
`snofall_tmp`, `snomelt_tmp`, `snomelt_max`, `snomelt_min`, `snomelt_lag`	`fall_tmp`, `melt_tmp`, `melt_max`, `melt_min`, `tmp_lag`, `snow.sno`	
`tile_dep`, `tile_dtime`, `tile_lag`, `tile_rad`, `tile_dist`, `tile_drain`, `tile_pump`	`dp`, `t_fc`, `lag`, `rad`, `dist`, `drain`, `pump`, `tiledrain.str`	
`tile_latk`	`lat_ksat`, `tiledrain.str`	The result is written to the tile lag time (`lag`).
`cmn_h`, `nperco_h`, `pperco_h`, `phoskd_h`, `psp_h`, `nperco_lchtile_h`	HRU values of the basin nutrient parameters `cmn`, `nperco`, `pperco`, `phoskd`, `psp` and `nperco_lchtile`	Change one HRU's copy. The basin names below change the basin value.

Soil layer parameters (`sol`)

Changed in soil layers `lyr1` to `lyr2` of each HRU.

Name	Input	Notes
z	layer depth dp, soils.sol	
bd, awc, k	bd, awc, soil_k, soils.sol	Dependent water contents are recomputed after awc changes.
clay, silt, sand, rock	clay, silt, sand, rock, soils.sol	
alb, usle_k, ec, cal, ph	alb, usle_k, ec, caco3, ph, soils.sol	
anion_excl, crk	anion_excl, perc_crk, soils.sol	Whole profile values.
cbn	carbon, soils.sol	No effect: the soil carbon pools are set up after the change list is applied.
lab_p, hum_c_n, hum_c_p	lab_p, hum_c_n, hum_c_p, nutrients.sol	No effect, for the same reason.

Plant parameters (plt)

Changed for the plants named in the row's plant conditions, in the plant community of each listed HRU. See Plant parameters for how to write these rows.

Name	Input
lai_pot	lai_pot, plants.plt
harv_idx	harv_idx, plants.plt
phu_mat	heat units to maturity of the plant in the HRU's community, computed at start-up from the plant and the climate

bio_e, hvsti, rdmx, t_opt, t_base and usle_c are recognised names, but rows for them change nothing.

Basin parameters (bsn)

Changed once for the whole basin.

Name	Input
surlag, prf, evrch, percop, cdn, sdnco, tb_adj, n_updis, rsd_covco	parameters.bsn
init_seq, init_microb, init_slow, init_passive, koc_c, solc_ratio, manure_c_frac, bio_consol, till_consol, sfc_rsd_photodeg, n_act_frac, cnr_cap, cnr_ref, cpr_cap, cpr_ref, t_cbn_min, t_cbn_opt, t_cbn_max	carbon.bsn
bmix_a, bmix_b, bmix_c, tillmix_a, tillmix_b, tillmix_c	biological and tillage mixing curve coefficients, carbon.bsn

Basin rows for adj_pkr, ffcb, dorm_hr, cmn, nperco, pperco, phoskd, psp, nperco_lchtile, rsdco, msk_co1, msk_co2, msk_x, plaps, tlaps and evlai change nothing. SWAT+ copies these basin values to the HRUs, channels, plants and weather stations while it sets them up, before the change list is applied, and does not read the basin value again. For the nutrient parameters, use the HRU names (cmn_h, nperco_h, pperco_h, phoskd_h, psp_h, nperco_lchtile_h) instead.

The per-layer carbon rates of carbon_lyr.bsn (hp_rate, hs_rate, microb_rate, meta_rate, str_rate, microb_top_rate, hs_hp, a1co2, asco2, apco2, abco2) are recognised names, but rows for them change nothing.

Aquifer parameters (aqu)

Name	Input
flo_init_mm, dep_bot, dep_wt_init, no3_init, minp_init, cbn_init, flo_dist, bf_max, alpha, revap_co, deep_seep, sp_yld, hlife_n, flo_min, revap_min	gw_flo, dep_bot, dep_wt, no3_n, sol_p, carbon, flo_dist, bf_max, alpha_bf, revap, rchg_dp, spec_yld, hl_no3n, flo_min, revap_min, aquifer.aqu

Channel parameters (`sdc` or `rte`)

Name	Input
`chw`, `chd`, `chs`, `chl`, `chn`, `chk`, `cov`, `d50`, `ch_clay`, `carbon`, `ch_bd`, `chss`, `bank_exp`, `vcr_coef`, `bankfull_flo`, `fps`, `fpn`, `hc_kh`, `hc_hgt`, `hc_ini`, `ch_n_conc`, `ch_p_conc`, `ch_p_bio`, `pk_rto`, `fp_inun_days`, `n_setl`, `p_setl`, `n_sol_part`, `p_sol_part`, `n_dep_enr`, `p_dep_enr`, `arc_len_fr`, `bed_exp`, `wash_bed_fr`	channel geometry, hydraulics, sediment and nutrient parameters of the SWAT-DEG channel inputs

Channel nutrient parameters (`swq`)

Name	Input
`mumax`, `rs1` to `rs7`, `rk1` to `rk6`, `bc1` to `bc4`	in-stream nutrient and algae rates, `nutrients.cha` (see channel inputs)

A `rk5` row changes `rs2` (the `ben_disp` column of `nutrients.cha`) instead.

Reservoir parameters (`res`)

Name	Input
`esa`, `evol`, `psa`, `pvol`	emergency and principal spillway surface areas and volumes, `hydrology.res`
`k_res`, `evrsv`	bottom hydraulic conductivity and evaporation coefficient, `hydrology.res`
`nsed`, `res_d50`, `sed_stlr`, `velsetlr`	normal sediment concentration, median particle size, settling rate and settling velocity, `sediment.res`
`psetlr1`, `psetlr2`, `nsetlr1`, `nsetlr2`, `nsolr`, `psolr`, `soln_stl_fr`, `solp_stl_fr`	nutrient settling rates, `nutrients.res`
`vol`, `sed`, `orgn`, `orgp`, `no3`, `nh3`, `no2`, `solp`	water, sediment and nutrient content at the start of the run, `initial.res`

Reservoir decision table parameters (`rdt`)

Name	Input
`drawdown_days`, `withdraw_rate`	release rules of the reservoir decision tables in `res_rel.dtl`, per storage zone (see Reservoir decision tables)

Pesticide parameters (`pst`)

List the pesticides to change (their order in `pesticide.pes`) in the object list. With an empty list nothing is changed.

Name	Input
`pst_koc`, `pst_washoff`, `pst_foliar_hlife`, `pst_soil_hlife`, `pst_solub`, `pst_aq_hlife`, `pst_aq_volat`, `pst_aq_settle`, `pst_aq_resus`, `pst_ben_hlife`, `pst_ben_bury`, `pst_ben_act_dep`	`soil_ads`, `frac_wash`, `hl_foliage`, `hl_soil`, `solub`, `aq_hlife`, `aq_volat`, `aq_settle`, `aq_resus`, `ben_hlife`, `ben_bury`, `ben_act_dep`, `pesticide.pes`

Simplified HRU parameters (`hlt`)

Name	Input
cn2_lte, awc_lte, etco_lte, tc_lte, soildep_lte, slope_lte, slopelen_lte, sy_lte, abf_lte, revapc_lte, percc_lte, sw_lte, gw_lte, gwflow_lte, gwdeep_lte, snow_lte, uslek_lte, uslec_lte, uslep_lte, uslels_lte	hru-lte.hru

awc_lte is given per mm of soil depth; SWAT+ multiplies the change and the upper limit by the soil depth.

gwflow parameters

These only act when gwflow is switched on in codes.bsn. Each name needs the object type shown.

Name	Object type	Objects changed	Input
aquifer_K, aquifer_Sy, aquifer_exdp	gwf	gwflow cells	aquifer hydraulic conductivity, specific yield and extinction depth, gwflow inputs
tile_K, floodplain_K	gwf	gwflow cells	tile drain and floodplain conductivity, only when tile or floodplain exchange is on
aquifer_delay	hru	HRUs	recharge delay from the soil to the water table, in days
stream_K, stream_thk	gwf_riv	channels	streambed conductivity and thickness of the cells connected to each channel
stream_bed	gwf_sgl	applied once	streambed elevation offset, one value for all channels
reservoir_K, pond_bed_K	gwf_sgl	applied once	reservoir and pond bed conductivity

Climate parameters (cli)

Name	Input
precip	daily precipitation of the listed *.pcp records, from day1 to day2 of each year from year1 to year2
temp	daily temperature of the listed *.tmp records; changes nothing

See Climate changes before using these.

Related pages

- Hard calibration. When and how the change list is applied.
- Conditions and regionalization. Condition line variables.
- Soft calibration. The seven .sft files that follow these two on the chg line.

Sources

- Readers: src/cal_parm_read.f90 and src/cal_parmchg_read.f90. Object list expansion: src/define_unit_elements.f90.
- Parameter names and the variables they change: src/cal_parm_select.f90. Object type dispatch: src/cal_conditions.f90.

- Types: `type calibration_parameters`, `type calibration_conditions` and `type update_parameters` in src/calibration_data_module.f90. File name slots: `in_chg%cal_parms` and `in_chg%cal_upd` in src/input_file_module.f90.

Column	Fortran variable
`cal_parms.cal` `name`, `obj_typ`, `abs_min`, `abs_max`, `units`	`cal_parms(i)%name`, `%ob_typ`, `%absmin`, `%absmax`, `%units`
`calibration.cal` `name` to `day2`	`cal_upd(i)%name`, `%chg_typ`, `%val`, `%conds`, `%lyr1`, `%lyr2`, `%year1`, `%year2`, `%day1`, `%day2`
`calibration.cal` `obj_tot` and list	`cal_upd(i)%num_tot`, expanded into `cal_upd(i)%num(:)`
condition `var alt targ targc`	`cal_upd(i)%cond(j)%var`, `%alt`, `%targ`, `%targc`
condition `range var min max`	`cal_upd(i)%cond(j)%var`, `cal_upd(i)%val1`, `cal_upd(i)%val2` (shared by all range lines of the row)